HyperLynx® SI/PI

DDR Simulation Report - IM-2400-runtwice-noCS-noEN-noMask-slowWeakCorners

Summary

Summary

- The simulated net is considered failed if at least one signal quality or timing requirement was not met.

Summary

#	Type	Number of Simulated Nets	Number of Failed Nets
1	Data Write	32	None
2	Data Read	32	None
3	Address	12	None
4	Differential Nets	6	None
5	CLK-DQS Skew	4	None

Data Write

Data Write Worstcases

- To investigate any result further, click on the link in the cell to open the waveforms. Viewing waveforms requires a valid Waveform Analyzer license.
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Data Write Worstcases

	Signal/Controller/DRAM			Status			Setup	Hold	Pulse Width	Min Slew Rate	Max Slew Rate	Voltage Margin between Signal and Eye-Mask2	Peak Voltage	Overshoot	Undershoot	Overshoot Area	Undershoot Area	
#	Signal	Driving Controller.Pin	Receiving DRAM.Pin	Pass/Fail	tUI [ps]	Reference voltage used [mV]	Margin [ps]	Margin [ps]	Margin [ps]	Margin [V/ns]	Margin [V/ns]	Margin [mV]	Above/Below Reference Voltage Margin [mV]	Margin [mV]	Margin [mV]	Margin [V*ns]	Margin [V*ns]	Monotonic
1	DDR4L_DQA[0]	U1.AD21	U3.B2	Pass	416.5	270.4	107.1	106.800	192.0	0.869	3.236	163.507	157.4	300.0	269.6	0.100	0.094	Pass
2	DDR4L_DQA[10]	U1.AE22	U3.E11	Pass	416.5	270.4	106.8	106.4	188.6	0.950	2.044	164.868	161.1	300.0	256.8	0.100	0.094	Pass
3	DDR4L_DQA[11]	U1.AD22	U3.F11	Pass	416.5	270.4	106.300	106.4	190.6	0.842	2.265	164.366	163.9	300.0	256.9	0.100	0.095	Pass
4	DDR4L_DQA[12]	U1.AH23	U3.F9	Pass	416.5	270.4	107.200	106.9	197.5	0.840	3.790	170.197	162.0	300.0	261.2	0.100	0.095	Pass
5	DDR4L_DQA[13]	U1.AH24	U3.E9	Pass	416.5	270.4	106.800	106.9	192.6	0.892	2.545	170.572	164.1	300.0	255.3	0.100	0.095	Pass
6	DDR4L_DQA[14]	U1.AE24	U3.C9	Pass	416.5	270.4	106.9	106.7	189.7	0.920	2.157	165.305	160.2	300.0	257.5	0.100	0.094	Pass
7	DDR4L_DQA[15]	U1.AG24	U3.B9	Pass	416.5	270.4	107.3	106.9	191.4	0.887	2.350	170.299	162.9	300.0	254.8	0.100	0.095	Pass
8	DDR4L_DQA[1]	U1.AE20	U3.C2	Pass	416.5	270.4	106.7	106.300	191.8	0.937	3.079	163.907	156.3	300.0	270.2	0.100	0.095	Pass
9	DDR4L_DQA[2]	U1.AD20	U3.E2	Pass	416.5	270.4	104.4	104.200	190.3	0.983	2.994	163.227	156.7	300.0	264.9	0.100	0.094	Pass
10	DDR4L_DQA[3]	U1.AF20	U3.F2	Pass	416.5	270.4	106.4	106.4	193.5	0.914	3.116	162.049	157.6	300.0	267.5	0.100	0.094	Pass
11	DDR4L_DQA[4]	U1.AH21	U3.F4	Pass	416.5	270.4	111.7	111.300	194.9	0.692	2.978	164.734	164.2	300.0	269.1	0.100	0.095	Pass
12	DDR4L_DQA[5]	U1.AH20	U3.E4	Pass	416.5	270.4	109.9	109.9	193.7	0.817	3.166	162.816	162.5	300.0	268.0	0.100	0.093	Pass
13	DDR4L_DQA[6]	U1.AH19	U3.C4	Pass	416.5	270.4	110.7	110.300	193.5	0.741	2.988	162.410	162.0	300.0	269.7	0.100	0.094	Pass
14	DDR4L_DQA[7]	U1.AG19	U3.B4	Pass	416.5	270.4	107.2	106.800	192.8	0.832	3.217	162.150	157.8	300.0	267.7	0.100	0.094	Pass
15	DDR4L_DQA[8]	U1.AF22	U3.B11	Pass	416.5	270.4	106.4	106.3	190.2	0.834	2.581	165.070	163.4	300.0	255.7	0.100	0.095	Pass
16	DDR4L_DQA[9]	U1.AH22	U3.C11	Pass	416.5	270.4	108.800	108.7	192.7	0.979	2.331	170.196	162.8	300.0	258.6	0.100	0.096	Pass
17	DDR4L_DQB[0]	U1.AC26	U3.AA2	Pass	416.5	270.4	112.4	112.4	198.1	0.657	2.873	168.511	165.0	300.0	270.7	0.100	0.094	Pass
18	DDR4L_DQB[10]	U1.AF28	U3.V11	Pass	416.5	270.4	114.800	114.7	193.3	0.778	2.800	168.130	169.6	300.0	266.5	0.100	0.094	Pass
19	DDR4L_DQB[11]	U1.AG28	U3.U11	Pass	416.5	270.4	113.7	113.300	196.9	0.732	2.625	169.271	169.0	300.0	266.9	0.100	0.095	Pass
20	DDR4L_DQB[12]	U1.AC27	U3.U9	Pass	416.5	270.4	105.4	105.4	193.9	0.904	2.948	169.188	160.3	300.0	259.9	0.100	0.095	Pass
21	DDR4L_DQB[13]	U1.AD27	U3.V9	Pass	416.5	270.4	110.8	110.800	194.0	0.824	3.124	165.060	164.2	300.0	269.4	0.100	0.095	Pass
22	DDR4L_DQB[14]	U1.AD28	U3.Y9	Pass	416.5	270.4	108.9	108.8	192.9	0.949	3.019	166.147	162.2	300.0	260.5	0.100	0.095	Pass
23	DDR4L_DQB[15]	U1.AC28	U3.AA9	Pass	416.5	270.4	105.7	105.300	192.1	0.922	3.093	169.720	165.3	300.0	263.3	0.100	0.094	Pass
24	DDR4L_DQB[1]	U1.AD26	U3.Y2	Pass	416.5	270.4	103.800	103.4	189.2	0.620	3.986	165.011	167.0	300.0	271.7	0.100	0.094	Pass
25	DDR4L_DQB[2]	U1.AD25	U3.V2	Pass	416.5	270.4	111.3	111.2	194.3	0.662	2.932	162.241	161.2	300.0	271.0	0.100	0.095	Pass
26	DDR4L_DQB[3]	U1.AD24	U3.U2	Pass	416.5	270.4	103.800	103.9	192.7	0.587	3.647	165.594	166.6	300.0	273.0	0.100	0.094	Pass
27	DDR4L_DQB[4]	U1.AG26	U3.U4	Pass	416.5	270.4	107.300	107.4	196.2	0.747	3.759	166.157	165.8	300.0	273.7	0.100	0.094	Pass
28	DDR4L_DQB[5]	U1.AH25	U3.V4	Pass	416.5	270.4	111.8	111.7	200.1	0.892	3.313	163.788	167.0	300.0	272.6	0.100	0.094	Pass
29	DDR4L_DQB[6]	U1.AH26	U3.Y4	Pass	416.5	270.4	112.9	112.8	197.5	0.903	3.201	166.367	166.0	300.0	274.6	0.100	0.095	Pass
30	DDR4L_DQB[7]	U1.AG25	U3.AA4	Pass	416.5	270.4	113.7	113.8	199.8	0.896	3.174	170.501	166.3	300.0	272.0	0.100	0.094	Pass
31	DDR4L_DQB[8]	U1.AH27	U3.AA11	Pass	416.5	270.4	106.300	106.300	190.8	0.869	2.872	170.336	171.5	300.0	254.4	0.100	0.095	Pass
32	DDR4L_DQB[9]	U1.AH28	U3.Y11	Pass	416.5	270.4	113.9	113.800	196.8	0.764	2.563	171.417	171.6	300.0	268.0	0.100	0.096	Pass

Data Read

Data Read Worstcases

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Data Read Worstcases

	Signal/DRAM/Controller			Status			Setup	Hold
#	Signal	Driving DRAM.Pin	Receiving Controller.Pin	Pass/Fail	tUI [ps]	Reference Voltage Used [mV]	Margin [ps]	Margin [ps]
1	DDR4L_DQA[0]	U3.B2	U1.AD21	Pass	416.5	179.8	89.3	89.3
2	DDR4L_DQA[10]	U3.E11	U1.AE22	Pass	416.5	181.1	88.8	90.1
3	DDR4L_DQA[11]	U3.F11	U1.AD22	Pass	416.5	181.1	89.2	89.9
4	DDR4L_DQA[12]	U3.F9	U1.AH23	Pass	416.5	181.1	89.8	89.9
5	DDR4L_DQA[13]	U3.E9	U1.AH24	Pass	416.5	181.1	88.2	89.3
6	DDR4L_DQA[14]	U3.C9	U1.AE24	Pass	416.5	181.1	89.3	89.9
7	DDR4L_DQA[15]	U3.B9	U1.AG24	Pass	416.5	181.1	88.8	90.2
8	DDR4L_DQA[1]	U3.C2	U1.AE20	Pass	416.5	179.8	88.8	88.7
9	DDR4L_DQA[2]	U3.E2	U1.AD20	Pass	416.5	179.8	88.3	88.3
10	DDR4L_DQA[3]	U3.F2	U1.AF20	Pass	416.5	179.8	89.3	89.0
11	DDR4L_DQA[4]	U3.F4	U1.AH21	Pass	416.5	179.8	87.3	87.1
12	DDR4L_DQA[5]	U3.E4	U1.AH20	Pass	416.5	179.8	88.5	88.5
13	DDR4L_DQA[6]	U3.C4	U1.AH19	Pass	416.5	179.8	87.5	87.4
14	DDR4L_DQA[7]	U3.B4	U1.AG19	Pass	416.5	179.8	89.5	89.6
15	DDR4L_DQA[8]	U3.B11	U1.AF22	Pass	416.5	181.1	89.5	90.3
16	DDR4L_DQA[9]	U3.C11	U1.AH22	Pass	416.5	181.1	88.6	89.7
17	DDR4L_DQB[0]	U3.AA2	U1.AC26	Pass	416.5	173.3	85.5	85.5
18	DDR4L_DQB[10]	U3.V11	U1.AF28	Pass	416.5	179.8	83.9	83.9
19	DDR4L_DQB[11]	U3.U11	U1.AG28	Pass	416.5	179.8	84.2	84.3
20	DDR4L_DQB[12]	U3.U9	U1.AC27	Pass	416.5	179.8	89.5	89.8
21	DDR4L_DQB[13]	U3.V9	U1.AD27	Pass	416.5	179.8	87.8	87.5
22	DDR4L_DQB[14]	U3.Y9	U1.AD28	Pass	416.5	179.8	88.5	88.8
23	DDR4L_DQB[15]	U3.AA9	U1.AC28	Pass	416.5	179.8	89.0	89.4
24	DDR4L_DQB[1]	U3.Y2	U1.AD26	Pass	416.5	173.3	86.3	86.1
25	DDR4L_DQB[2]	U3.V2	U1.AD25	Pass	416.5	173.3	87.2	87.3
26	DDR4L_DQB[3]	U3.U2	U1.AD24	Pass	416.5	173.3	84.4	89.2
27	DDR4L_DQB[4]	U3.U4	U1.AG26	Pass	416.5	173.3	85.1	84.9
28	DDR4L_DQB[5]	U3.V4	U1.AH25	Pass	416.5	173.3	82.9	82.7
29	DDR4L_DQB[6]	U3.Y4	U1.AH26	Pass	416.5	173.3	80.5	85.6
30	DDR4L_DQB[7]	U3.AA4	U1.AG25	Pass	416.5	173.3	84.5	84.5
31	DDR4L_DQB[8]	U3.AA11	U1.AH27	Pass	416.5	179.8	88.3	89.2
32	DDR4L_DQB[9]	U3.Y11	U1.AH28	Pass	416.5	179.8	82.3	82.9

Address

Address Worstcases

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Address Worstcases

	Signal/Dram		Status			Setup	Hold	Pulse Width	Min Slew Rate	Max Slew Rate	Voltage Margin between Signal and Eye-Mask	Peak Voltage	Overshoot	Undershoot	Overshoot Area	Undershoot Area	
#	Signal	Accessed DRAM	Pass/Fail	tUI [ps]	Reference voltage used [mV]	Margin [ps]	Margin [ps]	Margin [ps]	Margin [V/ns]	Margin [V/ns]	Margin [mV]	Above/Below Reference Voltage Margin [mV]	Margin [mV]	Margin [mV]	Margin [V*ns]	Margin [V*ns]	Monotonic
1	DDR4_CA[0]	U3.H2	**Pass**	833.0	274.3	228.7	254.3	348.3	1.089	0.957	175.668	169.6	300.0	278.8	0.100	0.098	Pass
2	DDR4_CA[1]	U3.J2	**Pass**	833.0	274.3	220.8	261.2	353.2	1.081	0.996	174.887	168.9	300.0	282.0	0.100	0.098	Pass
3	DDR4_CA[2]	U3.H9	**Pass**	833.0	274.3	214.100	267.5	354.1	1.073	1.382	174.528	174.4	300.0	282.8	0.100	0.098	Pass
4	DDR4_CA[3]	U3.H10	**Pass**	833.0	274.3	216.900	264.3	349.5	1.071	1.158	172.153	169.6	300.0	282.7	0.100	0.098	Pass
5	DDR4_CA[4]	U3.H11	**Pass**	833.0	274.3	226.7	254.5	349.5	1.048	0.943	173.941	169.4	300.0	280.1	0.100	0.098	Pass
6	DDR4_CA[5]	U3.J11	**Pass**	833.0	274.3	219.600	261.0	349.7	1.049	0.978	171.829	168.7	300.0	283.2	0.100	0.098	Pass
7	DDR4_CB[0]	U3.R2	**Pass**	833.0	274.3	204.900	273.0	354.1	0.975	1.924	172.763	170.4	300.0	284.1	0.100	0.096	Pass
8	DDR4_CB[1]	U3.P2	**Pass**	833.0	274.3	213.8	264.5	356.1	1.066	1.346	172.750	169.6	300.0	285.2	0.100	0.098	Pass
9	DDR4_CB[2]	U3.R9	**Pass**	833.0	274.3	193.7	280.4	353.8	0.890	2.232	164.910	169.5	300.0	286.3	0.100	0.098	Pass
10	DDR4_CB[3]	U3.R10	**Pass**	833.0	274.3	212.600	270.2	353.8	1.079	1.459	172.056	169.6	300.0	285.8	0.100	0.098	Pass
11	DDR4_CB[4]	U3.R11	**Pass**	833.0	274.3	195.7	275.6	353.4	0.908	2.299	166.574	169.6	300.0	285.5	0.100	0.098	Pass
12	DDR4_CB[5]	U3.P11	**Pass**	833.0	274.3	201.8	274.5	353.5	0.937	2.079	170.388	173.5	300.0	284.4	0.100	0.098	Pass

Differential Nets

Differential Nets Worstcases

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Differential Nets Worstcases

	Signal/Driver/Receicer				Status	Overshoot	Undershoot	Overshoot Area	Undershoot Area	Vix	Vinse Peak-to-peak	Vinse High	Vinse Low	
#	Signal	Driver.Pins	Receiver.Pins	Operation	Pass/Fail	Margin [mV]	Margin [mV]	Margin [V*ns]	Margin [V*ns]	Margin [mV]	Margin [mV]	Margin [mV]	Margin [mV]	Monotonic
1	DDR_CK_t_A	U1.W25&W26	U3.J8&J9	Write rank(1,1)	Pass	300.0	289.2	0.100	0.099	144.6	406.9	225.0	183.6	Pass
2	DDR_CK_t_B	U1.Y24&Y25	U3.P8&P9	Write rank(1,1)	Pass	300.0	295.1	0.100	0.100	134.6	402.2	220.2	179.6	Pass
3	PS_DDR_DQS_P0	U3.D3&E3	U1.AF21&AG21	Read rank(1,1)	Pass	N/A	N/A	N/A	N/A	N/A	167.4	89.0	80.5	N/A
4	PS_DDR_DQS_P0	U1.AF21&AG21	U3.D3&E3	Write rank(1,1)	Pass	300.0	291.4	0.100	0.100	114.2	385.7	212.8	178.0	Pass
5	PS_DDR_DQS_P1	U3.D10&E10	U1.AF23&AG23	Read rank(1,1)	Pass	N/A	N/A	N/A	N/A	N/A	173.1	91.2	85.3	N/A
6	PS_DDR_DQS_P1	U1.AF23&AG23	U3.D10&E10	Write rank(1,1)	Pass	300.0	291.3	0.100	0.100	109.1	403.0	226.8	175.0	Pass
7	PS_DDR_DQS_P2	U3.W3&V3	U1.AF25&AF26	Read rank(1,1)	Pass	N/A	N/A	N/A	N/A	N/A	172.9	100.5	77.1	N/A
8	PS_DDR_DQS_P2	U1.AF25&AF26	U3.W3&V3	Write rank(1,1)	Pass	300.0	292.8	0.100	0.100	114.5	403.6	218.0	180.3	Pass
9	PS_DDR_DQS_P3	U3.W10&V10	U1.AE27&AF27	Read rank(1,1)	Pass	N/A	N/A	N/A	N/A	N/A	184.0	100.4	86.2	N/A
10	PS_DDR_DQS_P3	U1.AE27&AF27	U3.W10&V10	Write rank(1,1)	Pass	300.0	292.0	0.100	0.100	113.0	400.3	220.0	175.5	Pass

CLK-DQS Skew

CLK-DQS Skew Worstcases

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CLK-DQS Skew Worstcases

	Signal/Dram				Status	DQSS - Earliest DQS	DQSS - Latest DQS	tDSS	tDSH
#	Signal	Accessed DRAM	DRAM CLK Pins	DRAM DQS Pins	Pass/Fail	Margin [ps]	Margin [ps]	Margin [ps]	Margin [ps]
1	PS_DDR_DQS_P0	U3	P8&P9	D3&E3	Pass	200.2	199.1	238.1	241.1
2	PS_DDR_DQS_P1	U3	P8&P9	D10&E10	Pass	200.2	200.0	239.2	240.8
3	PS_DDR_DQS_P2	U3	P8&P9	W3&V3	Pass	200.1	199.7	242.0	240.2
4	PS_DDR_DQS_P3	U3	P8&P9	W10&V10	Pass	200.2	200.0	241.5	240.7

Eye Density Links

Eye Density Links

Eye Density Links

#	Eye Density Links	Data/Address	Case
1	Load Eye Density Plots	Data	Slow
2	Load Eye Density Plots	Address	Slow

Setup Info

Design: 58500.100.01A.cce

24.09 build 25773879

Date: Oct-30-2026 9h-03m

Setup Information

Interface Setup

Parameter	Value
DDR Interface	LPDDR4
Crosstalk	Disabled
Power-Aware	Disabled
IC Corners	Slow
Probing location	Controller - at Die, DRAM - at Pin
Total Run Time	0 Days, 00 Hours, 02 Minutes 05 Seconds

Rank Setup

Rank	Ref. Des.	IBIS File	Component	Timing Model	Speed Grade
Controller	U1	fpga-xczu4eg.ibs	zynquplus	C:\SiemensEDA\HL2409\SDD_HOME\hyperlynx64\Libs\lpddr4_ctl.v	DDR_2400
Rank[1,1]	U3	lpddr4-IM.ibs	IME4G32L4HABG	C:\SiemensEDA\HL2409\SDD_HOME\hyperlynx64\Libs\lpddr4_dram.v	DDR_2400

ODT Setup Information

ODT - Write to Slot 1

Rank	DQS	DQ	DM
Controller	DWC_D5MQL4_40,	DWC_D5MPL4_40,	No Data
Rank[1,1]	DQ_RCV_ODT40_VOH3	DQ_RCV_ODT40_VOH3	No Data

ODT - Read from Slot 1

Rank	DQS	DQ	DM
Controller	DWC_D5MQL4_80ODT40,	DWC_D5MPL4_80ODT40,	No Data
Rank[1,1]	DQ_SOC40_PD40_VOH3	DQ_SOC40_PD40_VOH3	No Data